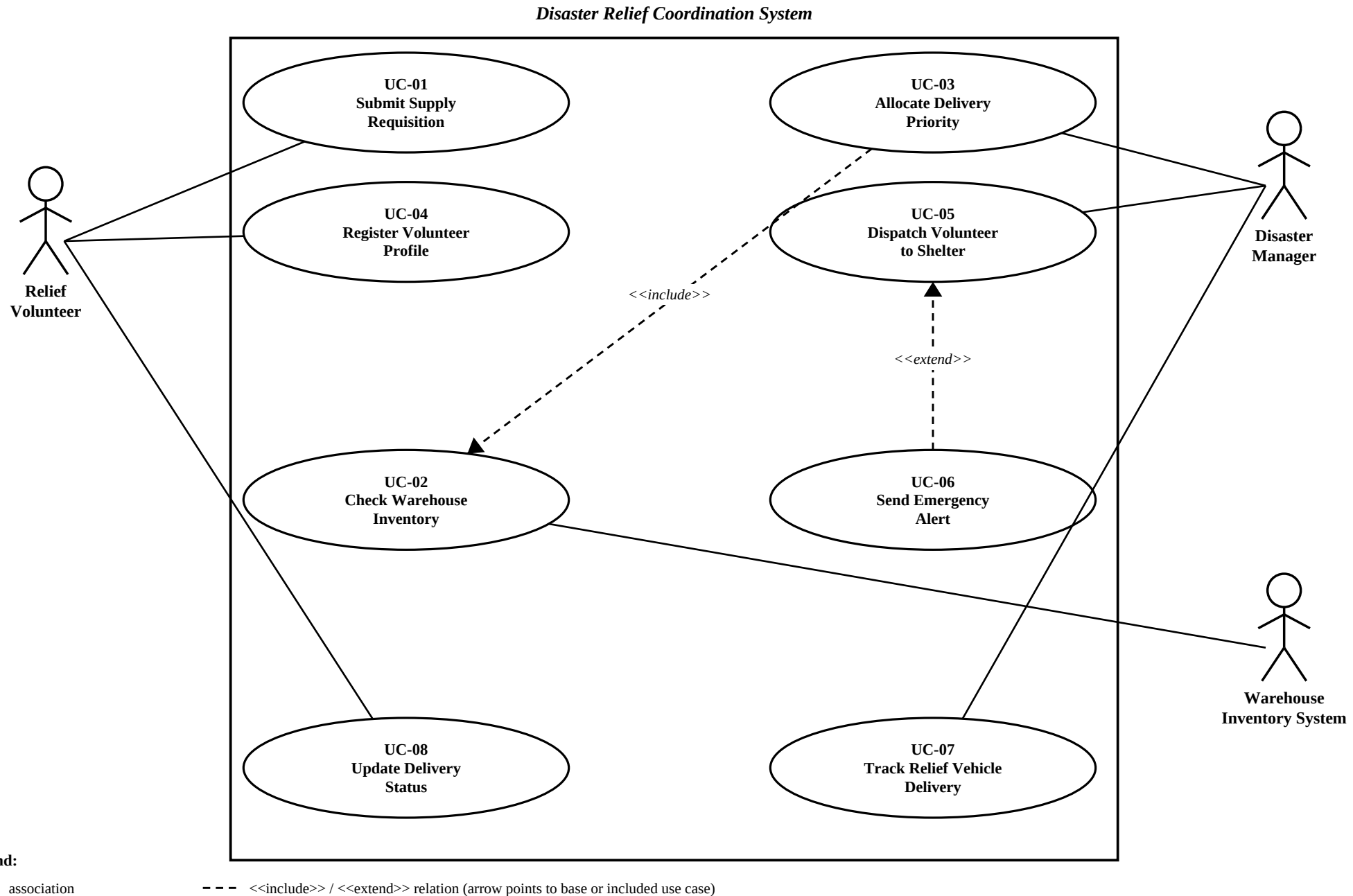


Requirements Table - Disaster Relief Supply & Volunteer Coordinator

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall match incoming field shelter supply requisitions against available central warehouse supplies and allocate delivery priority.	High	Pass: Critical need is given highest dispatch priority and routed to the logistics manifest. Fail: A low-priority item is dispatched ahead of a pending water/medical need.	This makes sure critical supplies like water and medical kits reach the shelters that need them most, instead of being sent out in whatever order requests come in.
FR-002	Functional	The system shall allow a Relief Volunteer to submit a field shelter supply requisition specifying item type, quantity, and urgency level.	High	Pass: A submitted requisition appears in the Disaster Manager's queue within 5 seconds with a unique requisition ID and all three fields populated. Fail: Requisition is not saved, or is missing item type, quantity, or urgency level.	All the later matching and dispatch decisions depend on this data being entered correctly at the start.
FR-003	Functional	The system shall allow a Relief Volunteer to register their skill set (e.g., medical, logistics, driving) and current availability status.	Medium	Pass: Volunteer profile is saved with at least one skill tag and an availability flag, and is searchable/filterable by the Disaster Manager. Fail: Profile saves without a skill tag or cannot be located in search.	Without this the Disaster Manager has no way to know which volunteer is actually qualified for a given task.
FR-004	Functional	The system shall allow a Disaster Manager to dispatch a matched, available Relief Volunteer to a specific shelter based on required skill and proximity.	High	Pass: Selected volunteer receives a dispatch notification (shelter location + task) within 10 seconds and volunteer status updates to "Assigned." Fail: Notification is not delivered, or status remains "Available."	This is the step that actually turns a matched need into a real assignment someone has to act on.
FR-005	Functional	The system shall allow a Disaster Manager or Relief Volunteer to track the real-time location and delivery status of a relief vehicle en route to a shelter.	Medium	Pass: Vehicle location refreshes at least every 60 seconds and each status change (Dispatched → In Transit → Delivered) is timestamped and visible. Fail: Location does not refresh within 60 seconds, or a status change is not logged.	Managers need to see where a delivery actually is, especially since conditions on the ground can change fast during a disaster.
NFR-001	Nonfunctional (Performance & Security)	The disaster mapping dashboard must operate with low network bandwidth consumption and support offline GIS data caching.	High	Pass: Benchmarking tests confirm target latency and security standards are met under simulated peak load.	Network connectivity in disaster zones is usually poor, so the app has to keep working even when the connection drops.
NFR-002	Nonfunctional (Security & Privacy)	The system shall encrypt all volunteer personal data and shelter location data in transit and at rest, and shall restrict data access by role (Relief Volunteer vs. Disaster Manager).	High	Pass: A security scan confirms TLS 1.2+ in transit and AES-256 at rest, and a Relief Volunteer account cannot view another volunteer's PII or the full warehouse inventory. Fail: Any unencrypted PII transfer/storage, or an unauthorized role successfully accesses restricted data.	Volunteer and shelter data is sensitive, so it needs to be protected even in the middle of an emergency response.

FR-001 and NFR-001 were given in the problem statement. FR-002 to FR-005 and NFR-002 were added to complete the set of five FRs and two NFRs.

Use Case Diagram - Disaster Relief Supply & Volunteer Coordinator



Use-Case Flow: Match Supply Requisition & Allocate Delivery Priority (UC-03)

Actors

Primary: Disaster Manager. Secondary: Warehouse Inventory System (checked automatically by the system).

Related Use Cases

<<include>> UC-02 Check Warehouse Inventory - this is always run as part of step 3 below.

Preconditions

- A field shelter supply requisition (UC-01) has been submitted and is in the pending queue.
- The Disaster Manager is authenticated and has dispatch/allocation permissions.
- The Warehouse Inventory System is reachable and its stock data is current.

Postconditions

- Success: The requisition is assigned a delivery priority level, reserved stock is deducted from warehouse inventory, and the requisition is added to the logistics manifest with status “Allocated.”
- Failure: The requisition is flagged “Insufficient Stock” or “On Hold,” no stock is reserved, and the Disaster Manager is notified.

Main Success Scenario

1. Disaster Manager opens the pending requisitions queue.
2. Disaster Manager selects a submitted requisition to process.
3. System retrieves the item type, quantity, and urgency level from the requisition and checks available central warehouse stock for the requested item(s) («include» UC-02 Check Warehouse Inventory).
4. System calculates a delivery priority score based on urgency level and item criticality (e.g., water and medical kits are weighted highest).
5. System reserves the required stock quantity against the requisition in the warehouse inventory.
6. System adds the requisition to the logistics manifest together with its assigned priority.
7. System updates the requisition status to “Allocated” and displays a confirmation to the Disaster Manager.
8. Use case ends successfully.

Alternate Flow - 4a. Insufficient Warehouse Stock

- 4a1. System detects that the requested quantity exceeds available warehouse stock.
- 4a2. System sets the requisition status to “Partial/Insufficient Stock” and suggests either a partial allocation or sourcing from a nearby warehouse, if available.
- 4a3. System notifies the Disaster Manager of the shortfall and the suggested options.
- 4a4. Disaster Manager confirms a partial allocation (flow resumes at step 5 with the reduced quantity) or places the requisition on hold pending restock (use case ends; requisition remains in the queue).

Notes

If the requisition is marked “Critical,” the Disaster Manager dispatching a Relief Volunteer (UC-05) afterward can optionally trigger Send Emergency Alert (UC-06) once this use case finishes.